

How Hyundai UX Team Develops

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ABSTRACT

More attention to automotive UX has risen due to Mobility as a service or Transportation as a service. The work of the team is to create a user experience design for clusters, an in-car infotainment system, a head up display, and a connected car smartphone app. Hyundai infotainment UX team provides various design case studies which show the user experience design of Ioniq5 EV vehicle, N UX design for high performance vehicle, connected car smartphone app, automotive voice UX, and the most recently launched Genesis GV60 infotainment system UX development process.

CCS CONCEPTS

• **Human-centered computing** → Interaction design; Interaction design process and methods; User interface design; Interaction design; Interaction design process and methods; User centered design; Interaction design; Interaction design process and methods; Scenario-based design; Human computer interaction (HCI); HCI design and evaluation methods; Usability testing.

KEYWORDS

Ioniq5, Hyundai N, GV60, Connected Car Service App

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1 INTRODUCTION

Nowadays, more attention to an automotive UX has arisen continuously due to Mobility as a service and Transportation as a service after automotive UX becomes one of key solutions for seamless connectivity.

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Figure 1: Products Hyundai UX team develops.

It is not commonly exposed to the public about how automotive UX has been researched and developed because automobile industries are somewhat conservative. Therefore, the infotainment UX development team who is in charge of in-car infotainment system UX in Hyundai will share the case studies based on the mass production.

The work includes clusters, in-car infotainment systems, HUD (head up display), and smartphone apps which mainly provide the control over vehicles remotely and connect maintenance services. The team designs and develops technical UX specifications based on drivers and passengers' behaviors, technical issues and regulations.

Practical guidelines are needed for designing the automotive UX to allocate a variety of information because drivers interact with numbers of displays, physical interfaces, and road information. In addition, climate and seat control interface helps users to reduce tiredness and to enjoy entertainment with a better driving condition.

The regulation of safety and local policies are highly strict and important factors for designing UX. For instance, NHTSA driving distraction guides have an influence over design components such as the scale of cluster gauges, duration of seat belt alarms, visibility of a defrost switch, and OTA(over-to-air) condition.

The workshop consists of a visual design of 1) Ioniq 5 EV vehicle which requires to consider not only its brand but also interior character lines and assets because the vehicle is built on a Hyundai's new EV platform, 2) Hyundai Veloster N, a vehicle for track driving, has recently received new designs and features such as a lap timer and a circuit. 3) Smartphone apps provide control over vehicles



Figure 2: IONIQ5 Cluster and Infotainment

and it's also compelling to be a center of connected car services as a super app. 4) Voice UX especially focuses on completing the tasks for vehicles. 5) Genesis GV60 infotainment system offers a new experience for the premium brand.

2 CASE STUDIES

Following design cases provide sample design sketches and development processes done in recent works.

2.1 Visual-centric Infotainment UX Design Case

It explains the background and development process of creating unique graphics for each brand, focusing on actual cases.

2.1.1 IONIQ 5. IONIQ 5 is the first electric vehicle equipped with Hyundai's dedicated EV platform, E-GMP. As users expect new experiences for the new electric vehicle platform, we have developed it from the scratch. We have discovered paper metaphors that go well with clean and pure feeling of electric cars. The point color that goes from blue to green is applied to give pure energy of the electric vehicle. In addition, character elements of the vehicle's interior and exterior design were applied to the graphics to enhance the overall sense of harmony. Through this new concept, we've won many Global design awards and contributed to the selection of IONIQ 5 as the World Car of the Year.

2.1.2 Hyundai Aqua, Kia Magenta theme. Hyundai starts with the color of blue. We have discovered the Aqua color shown in the deep sea and emphasized the futuristic feeling with the isometric cube shape. In the case of Kia, we discovered the most comfortable magenta color on the infotainment screen in the red color spectrum, and we were able to emphasize the dynamic and young brand image using neon images. In addition, the world's first cluster theme responding to weather, time, and driving speed was applied to Kia. This responsive cluster design provides users with pleasure and comfort. As a result, currently, the Aqua and Magenta themes have received good responses from domestic and international customers. Lastly, the future platform will further provide an integrated visual design including brands, interior and exterior design of vehicles.

2.2 Hyundai N

Recent designs and feature updates made in Hyundai N are explained as a case study.

2.2.1 Research case study. Introduce research and analysis on information obtained from interviews with racing drivers to find out which information is imperative for driving and how it should be prioritized.

Share recent results of in-depth interviews of race track drivers and insights found.

Discuss what benefits users can earn from the updated N Mode design.

2.2.2 Design process case study. Recent design update of Veloster N is presented with a design process. "Circuit" and "Lap time" are new features in Veloster N and share the process of how previous designs have advanced. Besides, explain how these new features help the racing experience to be continued even after the race is finished.

2.3 Mobile UX Development for Connected Car Service

2.3.1 A study on the motor sports market and customer needs through the Hyundai N track app. Investigate the background of the N track app inspired by motor sports and the usage and needs of high-performance vehicle owners, and explain how the customer's experience is designed from a UX perspective (e.g., Gamification).

Share what types of analysis function and the N track app which improves driving skills. Introduce how instructors can use them as educational tools.

2.3.2 The present and future of Hyundai/Genesis/Kia Connected Car Service. Introduce key features and design characteristics of the current CCS app UX 3.0, share environmental limitations, overcome history, and results.

Introduce the vision of a super app that can include the overall mobility experience of customers beyond the existing service scope that was focused on vehicle control, monitoring and connection with external services.

Discuss the concerns in the aspect of UX design and benefits that customers can earned from CCS app UX 4.0.

2.4 Automotive Voice UX Development

Introduce the Automotive Voice UX and differentiate Automotive Voice UX from AI Speakers

Also, introduce a scenario of setting a destination with "Find Starbucks" and discuss key features such as Display, Input Stages, and Status Recognition.

Share insights found from field issues, recently updated Voice UX design, and its new features with motion.

2.5 Connected Car Integrated Cockpit UX s

2.5.1 ccIC UX. The connected car Integrated Cockpit (ccIC) UI has been applied to the new infotainment system in Genesis cars for the first time. The ccIC platform has been designed with a focus on smooth content linkage between HUD, cluster and AVNT (Navigation) while providing an integrated user experience. UI features present on all screens communicate Genesis' coherent brand philosophy. By providing information the driver ought to know under appropriate circumstances, with the optimal method, supports a safe and comfortable driving experience. ccIC has inherited and



Figure 3: GENESIS ccIC Cluster and Infotainment

reinterpreted the advantages of the existing skeuomorphism, flat and neumorphism designs, hereby proving a grand experience for both customers who desire the dignified beauty of the Genesis brand and those who desire eco-friendly novelty.

2.5.2 Separated and Wide Displays. Within the scope of the interior design, it is difficult to disregard the physical distance between

the two separately configured displays. Nonetheless, efforts to embrace the physical obstacles for the space between displays to seem as if it were part of the design from the perspective of the driver. These were carried out by using the spatial distance as the role of a pillar in an animation where a vehicle is passing behind, or by making the impression of a shade existing in front of a connected background. In relation to the wide display, ccIC UX attempted to differentiate itself by providing real image content over the simple but intuitive card-form UI components which flow in the horizontal direction but don't divide the screen into narrow height.

2.5.3 Light Theme of Graphics. Firm establishment of the grounding atmosphere has been carried out primarily to construct all the graphic elements under a single design philosophy. Two graphic themes have been provided to consider individual preferences and an automatical change of themes during the sunset hour from white to black has taken both convenience and preference into consideration at the same time. Encompassing all these innovative functions, Genesis is constructing distinguished usability and aesthetics unique to the brand.